

EDUCATION

Tufts University <i>MS in Mathematics</i> <i>BS in Chemical Engineering; GPA: 3.9</i> Second Major: <i>Mathematics</i> ; Minor: <i>Chemical Physics</i> Thesis (Highest Honors): <i>Data Augmentation Models of Stem Cell Growth for Machine Learning Architecture Design</i> Committee: Emmanuel Tzanakakis (Advisor), James M. Murphy, Nathaniel Eagan	Medford, MA <i>Sep 2025 - May 2026</i> <i>Sep 2021 - May 2025</i>
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ACADEMIC PAPERS

Augmentation models of stem cell culture data for machine learning <i>Owen Skriloff, Demetrios Stoukides, Emmanuel Tzanakakis</i> - Built and trained three physics informed ML models of stem cell growth dynamics with PyTorch and SciPy. - Built novel data pipeline with Monte Carlo simulation and feature importance to assess model alignment with biology.	Biotechnology and Bioengineering <i>Submitted Jun 2025</i>
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ACADEMIC PROJECTS

Connecting loss-landscape geometry to generalization across MLP depths <i>Owen Skriloff (for Usman Khan)</i> - Investigated how network depth affects generalization by analyzing loss landscape curvature - Identified optimal depth and improved validation error by 15% using Hessian-based regularization.	Iterative Methods in ML <i>May 2025</i>
Process design and economic analysis of an industrial amikacin synthesis plant <i>Owen Skriloff, David Gillingham, Ivy Le, Younseo Yi</i> - Designed and simulated industrial scale process including bioprocessing and chemical reactive steps using SuperPro - Performed Class 4 economic analysis of plant profitability using labor, material, and equipment costs	CHBE Capstone <i>Apr 2025</i>
Energy and Entropy Measures for Reaction Diffusion PDE Dynamic Analysis <i>Owen Skriloff (for James Adler)</i> - Built and proved convergence of finite-element numerical solver in Python for reaction diffusion systems. - Analyzed dynamics under varied conditions using entropy and energy measures.	Numerical PDEs <i>Dec 2024</i>
Quantifying stress induced mineralization with a $BaTiO_3$ piezoelectric composite <i>Owen Skriloff (advised by Santiago Orrego)</i> Developed and performed methodology to quantify a relationship between stress and mineral deposits generated by electric charges from a piezoelectric composite using simulated body fluid, incubator, and actuator.	Temple University <i>Aug 2019</i>

INDUSTRY EXPERIENCE

Itaconix Corporation <i>Summer Associate Engineer</i> - Commissioned, purchased parts, and assembled a 100L reactor and scaled an esterification process from 250mL to 50L. - Developed HPLC and GPC methods for 200+ samples and supported GPC installation.	Stratham, NH <i>Jun 2023 - Aug 2023</i>
Araca Inc <i>Laboratory Technician</i> - Conducted 25+ chemical mechanical planarization (CMP) and post-CMP cleaning experiments in a Class 100 cleanroom. - Measured 250+ wafers, mixed slurries, programmed polishers, and tested new CMP products. - Analyzed data in MATLAB, Excel, and LabView to create 25+ presentations supporting client process improvements.	Tucson, AZ <i>Sep 2020 - Dec 2020</i>

CONFERENCES

AICHE Annual Meeting <i>Speaker: Data augmentation for ML architecture design</i>	Boston, MA <i>Nov 2025</i>
Cell Culture Engineering XIX Conference <i>Poster: Data augmentation for stem cell bioprocess understanding</i>	Tucson, AZ <i>Apr 2025</i>

HONORS AND AWARDS

Benjamin G Brown Scholarship (Tufts, 2025); Christos Georgakis Award (Tufts, 2025); Summa Cum Laude (Tufts, 2025); Highest Thesis Honors (Tufts, 2025); Summer Undergraduate Research Grant (Tufts, 2024); Regeneron Science Talent Search Top Scholar (2020); National Science and Humanities Symposium: 3rd Place Physical Science (2020).

SKILLS

Languages: Python, MATLAB	Tools: PyTorch/TensorFlow, SciPy, Numpy/Pandas, Matplotlib/Seaborn, Git
Mathematical Skills: Optimization, algorithmic and numerical analysis, high dimensional data analytics, data visualization, OOP, time series data analysis, finite element method.	
Chemical Engineering Skills: Reaction design and scale up, chemstation, HPLC and GPC, chemical mechanical planarization, reaction kinetic modeling, process control modeling, SuperPro.	